Monitor the Health of Your System with New Relic

Learn how New Relic can be effectively used to monitor Dockerized applications in a production environment. Walk through a step-by-step integration, which explains key monitoring metrics and demonstrates how organisations can leverage New Relic's alerting, logging, and visualisation capabilities to maintain optimal system performance.

In the era of cloud-native applications and microservices, efficient monitoring is essential for ensuring performance, availability, and reliability. As businesses scale their applications using Docker and Kubernetes, traditional monitoring techniques struggle to provide deep visibility into containerised workloads, distributed applications, and dynamic infrastructures. This calls for robust observability solutions that offer real-time insights, proactive alerting, and performance diagnostics.

New Relic, a powerful application performance monitoring (APM) tool, addresses this need by providing end-to-end observability across containers, microservices, and infrastructure components. It seamlessly integrates with Dockerized environments, offering deep insights into application behaviour, resource utilisation, network performance, and security vulnerabilities.

Key features of New Relic

New Relic is a comprehensive observability platform particularly valuable for containerised environments like Docker and Kubernetes. Unlike traditional monitoring solutions that focus only on basic CPU, memory, and network metrics, New Relic delivers a holistic view of system health by integrating application performance monitoring (APM), distributed tracing, infrastructure monitoring, and intelligent alerting into a single platform. This enables DevOps and site reliability engineering (SRE) teams to detect issues proactively, optimise resource utilisation, and prevent downtime in microservices-based architectures. Here are its key features.

Real-time application performance monitoring (APM): Continuously tracks application response times, error rates, and transaction throughput to detect performance bottlenecks.

Infrastructure monitoring: Provides deep visibility into containerised workloads, virtual machines, cloud platforms (AWS, Azure, GCP), and on-premises infrastructure.

Distributed tracing: Traces end-to-end user requests across microservices to help pinpoint slow dependencies, failures, and latency issues.

Logs in context: Unifies logs, metrics, and traces in a single view, enabling efficient root cause analysis and troubleshooting.

Intelligent alerts and anomaly detection: Uses machine learning-powered insights to detect anomalies, proactively triggering alerts before issues escalate.

Custom dashboards and visualisation: Provides interactive dashboards that allow teams to monitor key performance indicators (KPIs) and system trends in real time.

Auto-instrumentation and OpenTelemetry support: Seamlessly integrates with open-source observability tools like Prometheus, Grafana, and OpenTelemetry, ensuring